PROJECT DESIGN DOCUMENT

End-to-End Medallion Architecture

Healthcare & Hospital Management System

Data Source	Architecture	Platform
Flat CSV Files (5 Sources)	Medallion (Bronze→Silver→Gold)	Databricks / Azure Data Lake

Domain: Healthcare | *Compliance: HIPAA* | *Date: April 2026* | *Version: 1.0*

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Data Sensitivity Level	HIGH — Contains Patient Health Data
Business Domain	Healthcare Analytics & Hospital Operations
Project Sponsor	VP of Data & Analytics
Project Manager	Data Engineering Lead
Target Platform	Azure Databricks + Azure Data Lake Storage Gen2
Pipeline Pattern	Medallion Architecture (Bronze / Silver / Gold)

Metadata Field	Details
Ingestion Method	Metadata-Driven Framework (Delta-based config table)
Scheduling	Databricks Workflows — Daily Batch
SLA / Freshness	Data available by 06:00 AM daily
Data Retention Policy	Bronze: 7 years (HIPAA) | Silver: 5 years | Gold: 3 years
Backup & DR	Azure Geo-Redundant Storage (GRS)
Access Control Model	RBAC via Unity Catalog (Column-level security)
PII Handling	SHA-256 hashing for SSN, DOB, Phone in Silver layer
Related Documents	Data Dictionary v1.0, HIPAA Risk Assessment, Source System Specs
Change Control Process	JIRA Ticket → Peer Review → Arch Board Approval
Distribution List	Data Engineering, BI Team, Compliance Officer, CTO Office
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Revision History

Version	Date	Author	Change Summary	Approved By
1.0	April 2026	Data Eng. Team	Initial document creation — full architecture design	Arch. Review Board
1.1	TBD	TBD	Post-review updates and feedback incorporation	TBD
2.0	TBD	TBD	Production-ready version with Unity Catalog integration	CTO Office

1. Problem Statement

Hospital networks today operate with patient data scattered across multiple departments — each maintaining its own isolated systems with no unified data pipeline. This fragmentation leads to missed insights, billing inefficiencies, and poor patient outcomes.

1.1 Data Domains & Source Overview

Data Domain	Source Format	Key Information
Patient Master	Flat CSV Files	Demographics, medical history, insurance details
Appointment Data	Flat CSV Files	Doctor schedules, patient visits, duration
Billing & Claims	Flat CSV Files	Patient charges, insurance claims, payments
Lab Results	Flat CSV Files	Test results, abnormal flags, trends
Medication / Pharmacy	Flat CSV Files	Prescriptions, drug inventory, costs

1.2 Business Challenges

- **Patient data is scattered across departments with manual record-keeping**
- No single source of truth for patient demographics or history
- Data is siloed — billing has no visibility into lab results or medications

- **Cannot predict patient no-shows**
- Revenue loss due to empty appointment slots
- No historical pattern analysis on no-show behavior

- **Billing delays and claim rejections**
- Manual processes cause errors in charge codes
- Insurance claim approval rates are untracked

- **Poor medication inventory management**
- Overstocking or stockouts of critical medications
- No turnover analysis or expiry tracking

- **Cannot identify patients at risk of chronic diseases**
- Lab trends go unanalyzed across visits
- No readmission prediction or early intervention capability

2. What This Project Will Solve

By implementing an end-to-end Medallion Architecture on a Lakehouse platform, we will build a centralized, governed, and analytics-ready data platform that solves the following:

Problem	Solution Delivered
Scattered patient data	Unified Patient 360 view in Silver layer joining all 5 data sources
No-show prediction	Historical no-show rate KPI + trend tracking in Gold layer
Billing delays	Billing accuracy rate and claim approval rate KPIs with data lineage
Medication waste	Medication inventory turnover tracked daily in Gold layer
Chronic disease risk	Lab result trends and readmission rates surfaced as Gold KPIs
No audit trail	Full audit log capturing every pipeline run, row count, and error
Manual & error-prone	Metadata-driven ingestion framework — zero hardcoded logic
HIPAA compliance gap	Data governance columns, audit logs, and access-controlled layers

3. Prerequisites

3.1 Platform & Infrastructure

- Databricks Workspace (Azure Databricks or Community Edition)
- Azure Data Lake Storage Gen2 (ADLS Gen2) or Databricks File System (DBFS)
- Apache Spark 3.x runtime (Databricks Runtime 12.x or higher)
- Delta Lake enabled (included in Databricks by default)

3.2 Access & Permissions

- Storage account access key or Service Principal with Storage Blob Data Contributor role
- Databricks cluster with at least 2 worker nodes (Standard_DS3_v2 recommended)
- Unity Catalog enabled (optional but recommended for governance)

3.3 Tools & Libraries

Tool / Library	Version	Purpose
PySpark	3.x (built-in)	Core data processing engine
Delta Lake	2.x (built-in)	ACID transactions on Bronze/Silver/Gold
Python	3.9+	Scripting and utility functions
pandas	1.5+	Metadata config management
Great Expectations (optional)	0.17+	Data quality validation in Silver
Databricks SQL	Built-in	Gold layer querying and KPI dashboards

3.4 Input Data Files

Five CSV source files placed in ADLS/DBFS landing zone:

- patient_master.csv
- appointments.csv
- billing_claims.csv
- lab_results.csv
- medications.csv
- Metadata configuration file (or Delta table) defining schema, file path, and target layer for each source

4. Architecture Overview

The project follows the Medallion Architecture pattern — a layered approach to building reliable, scalable data pipelines on a Lakehouse platform.

🥉 BRONZE LAYER	🥈 SILVER LAYER	🥇 GOLD LAYER
Raw Ingestion	*Cleansed & Enriched*	*KPIs & Analytics*
• CSV files ingested as-is • No transformation • Metadata columns added • Audit log written • Delta format	• Null / duplicate removal • Schema standardization • Business rule validation • Cross-source joins • SCD Type 2 (patients)	• Aggregated KPI tables • Business metrics • Dashboard-ready data • Scheduled refresh • Serves BI / ML teams

5. Layer-by-Layer Detail

5.1 Bronze Layer — Raw Ingestion

The Bronze layer is the landing zone. Data is ingested exactly as received from source CSV files with zero business transformation. The purpose is to preserve the original record for lineage and reprocessing.

Audit & System Metadata Columns (Bronze)

Column Name	Data Type	Description	Category
_ingestion_timestamp	TIMESTAMP	Exact datetime the record was loaded into Bronze	Audit
_source_file_name	STRING	Name of the CSV file that produced the row	Lineage
_batch_id	STRING	Unique ID for each pipeline run (UUID)	Audit
_layer	STRING	Constant value ‘BRONZE’ for layer identification	Governance
_record_hash	STRING	MD5/SHA hash of full row for change detection	Quality
_ingestion_date	DATE	Partition column — date portion of ingestion timestamp	Partition
_pipeline_version	STRING	Version of the ingestion notebook that loaded the record	Lineage
_source_system	STRING	Originating system identifier (e.g., EHR, LIS, PMS)	Lineage
_is_duplicate	BOOLEAN	Flag set in Bronze if duplicate row detected at ingest	Quality
_raw_row_number	LONG	Original row position in source CSV file	Lineage

5.2 Silver Layer — Cleansed & Enriched

The Silver layer transforms raw Bronze data into reliable, business-validated datasets. Cross-source enrichment (joins) also happens here.

Data Quality & Governance Metadata Columns (Silver)

Column Name	Data Type	Description	Category
_dq_passed	BOOLEAN	True if record passed all data quality checks	Quality
_dq_failure_reason	STRING	Comma-separated list of DQ rules that failed	Quality

Column Name	Data Type	Description	Category
_dq_score	FLOAT	Numeric score 0.0–1.0 representing data completeness	Quality
_silver_load_timestamp	TIMESTAMP	When the record was written to Silver	Audit
_silver_batch_id	STRING	Batch ID of the Silver pipeline run	Audit
_bronze_batch_id	STRING	Reference back to source Bronze batch ID	Lineage
_is_current	BOOLEAN	SCD Type 2 flag — True for current active version	SCD
_effective_from	TIMESTAMP	SCD Type 2 — when this version became active	SCD
_effective_to	TIMESTAMP	SCD Type 2 — when this version was superseded	SCD
_masked_fields	STRING	Comma-separated list of PII fields that were masked	Privacy
_enrichment_source	STRING	List of source tables used in cross-source join	Lineage
_record_version	INT	Incremental version counter per primary key	SCD

5.3 Gold Layer — KPIs & Business Metrics

The Gold layer is the analytics-serving layer. Aggregated, denormalized tables are built here — purpose-built for dashboards, reports, and KPI tracking.

KPI Reporting Metadata Columns (Gold)

Column Name	Data Type	Description	Category
_gold_load_timestamp	TIMESTAMP	When the KPI record was written to Gold	Audit
_gold_batch_id	STRING	Batch ID for traceability from Gold back to Silver	Lineage
_kpi_period	STRING	Reporting period: DAILY / WEEKLY / MONTHLY	Reporting
_kpi_version	INT	Restatement counter — increments on recalculation	Versioning
_silver_source_tables	STRING	List of Silver tables that fed this KPI	Lineage
_department_id	STRING	Hospital department or cost centre identifier	Reporting
_report_as_of_date	DATE	Business date this KPI snapshot represents	Reporting

Column Name	Data Type	Description	Category
_threshold_status	STRING	GREEN / AMBER / RED based on KPI threshold	Alerting
_is_restatement	BOOLEAN	True if this record is a corrected restatement	Quality

KPI Tables in Gold

KPI	Formula / Logic	Source Tables (Silver)
Patient No-Show Rate	No-shows / Total Appts × 100	silver_appointments
Avg Length of Stay (ALOS)	SUM(discharge − admit) / COUNT(patients)	silver_appointments + silver_patients
HCAHPS Satisfaction Score	AVG(patient_satisfaction_score)	silver_appointments
Billing Accuracy Rate	Accurate Bills / Total Bills × 100	silver_billing
Insurance Claim Approval Rate	Approved Claims / Total Claims × 100	silver_billing
Medication Inventory Turnover	Units Dispensed / Avg Inventory Units	silver_medications
Operating Cost per Patient	Total Operating Cost / Unique Patients	silver_billing + silver_medications
Patient Readmission Rate	Readmissions within 30 days / Discharges × 100	silver_appointments

6. Metadata-Driven Ingestion Framework

Instead of hardcoding file paths and schemas in each notebook, a central Metadata Configuration Table (Delta table) drives all ingestion automatically. This makes the pipeline reusable, auditable, and easy to extend.

6.1 Metadata Config Table — Full Schema

Column	Data Type	Example Value	Purpose	Added
source_id	STRING	SRC_001	Unique ID for each data source	Original
source_name	STRING	patient_master	Logical name of the source	Original
file_path	STRING	/mnt/landing/patient_master.csv	Full path to the CSV file	Original
file_format	STRING	csv	Format of the source file	Original
target_table	STRING	bronze.patient_master	Delta table to load into	Original
primary_key	STRING	patient_id	Primary key for dedup in Silver	Original
active_flag	STRING	Y	Y = process this source, N = skip	Original
load_type	STRING	FULL / INCREMENTAL	Load strategy	Original
last_load_timestamp	TIMESTAMP	2026-04-15 10:00:00	Timestamp of last successful load	Original
expected_schema_version	STRING	v1.2	Schema version for validation against source	New
source_system_owner	STRING	EHR Team	Team responsible for the upstream source	New
data_classification	STRING	PHI / PII / INTERNAL	HIPAA sensitivity classification	New
row_count_threshold	LONG	1000	Minimum expected rows; alert if below threshold	New
max_null_pct	FLOAT	0.05	Max allowed null % for mandatory fields (e.g., 5%)	New
notification_email	STRING	data-ops@hospital.org	Alert email if pipeline fails for this source	New
retry_count	INT	3	Number of retry attempts on transient failure	New
quarantine_path	STRING	/mnt/quarantine/patient_master/	Path to write rejected / bad records	New
partition_column	STRING	ingestion_date	Column used to partition the Bronze Delta table	New

Column	Data Type	Example Value	Purpose	Added
sla_cutoff_time	STRING	06:00 AM UTC	Deadline by which this source must be loaded	New
dependency_source_ids	STRING	SRC_001,SRC_002	Comma-separated IDs of upstream dependencies	New

6.2 How It Works

- At pipeline start, the orchestration notebook reads the metadata config table
- For each active source (active_flag = Y), it triggers the Bronze ingest notebook
- The same notebook works for all 5 sources — no duplication of code
- On successful load, last_load_timestamp is updated in the config table
- Failed sources are logged in the audit log without stopping other sources

7. Audit Log — Design & Purpose

Every pipeline execution is tracked in a central Audit Log Delta table. This ensures full traceability, supports debugging, and satisfies HIPAA compliance requirements around data access and modification tracking.

7.1 Audit Log Table Schema — Full (with New Columns)

Column	Data Type	Description	Added
audit_id	STRING (UUID)	Unique identifier for each audit entry	Original
batch_id	STRING	Groups all tables loaded in a single pipeline run	Original
source_name	STRING	Name of the data source / table processed	Original
layer	STRING	BRONZE / SILVER / GOLD	Original
pipeline_start_time	TIMESTAMP	When this layer/source started processing	Original
pipeline_end_time	TIMESTAMP	When processing completed	Original
rows_read	LONG	Total rows read from source	Original
rows_written	LONG	Rows successfully written to target	Original
rows_rejected	LONG	Rows that failed validation (Silver)	Original
status	STRING	SUCCESS / FAILED / PARTIAL	Original
error_message	STRING	Error details if status = FAILED	Original
triggered_by	STRING	User or scheduler that triggered the run	Original
created_at	TIMESTAMP	Timestamp this audit record was created	Original
pipeline_duration_secs	LONG	Total execution time in seconds for SLA monitoring	New
notebook_name	STRING	Name of the Databricks notebook that ran	New
cluster_id	STRING	Databricks cluster ID used for the run	New
spark_app_id	STRING	Spark application ID for cross-referencing logs	New
rows_quarantined	LONG	Rows sent to quarantine path (bad/malformed records)	New
dq_score_avg	FLOAT	Average data quality score across all rows in this batch	New
schema_version	STRING	Schema version used during this pipeline run	New
environment	STRING	DEV / UAT / PROD environment identifier	New
retry_attempt	INT	Retry attempt number (0 = first attempt)	New

Column	Data Type	Description	Added
data_classification	STRING	PHI / PII / INTERNAL — copied from metadata config	New
sla_met	BOOLEAN	True if pipeline completed before sla_cutoff_time	New
downstream_notified	BOOLEAN	True if downstream teams were notified on completion	New

7.2 What the Audit Log Enables

- Full pipeline lineage — trace any Gold KPI back to its Bronze source row
- Failure investigation — quickly identify which source failed and why
- SLA monitoring — track pipeline duration trends over time using pipeline_duration_secs
- HIPAA compliance — maintain an immutable record of all data processing activity
- Data quality trend — monitor rows_rejected and dq_score_avg over time
- Environment segregation — differentiate DEV / UAT / PROD runs in one log table

8. Notebook Plan

The project will be implemented across 8 Databricks notebooks, organized by layer and responsibility.

#	Notebook Name	Layer	Purpose
00	setup_config	Setup	Create metadata config table, mount ADLS, initialize audit log table, set global variables
01	ingest_bronze	Bronze	Metadata-driven CSV ingestion for all 5 sources. Appends audit columns. Writes to Delta. Logs to audit.
02	silver_patients	Silver	Cleanse Patient Master: nulls, dedup, SCD Type 2, insurance validation. Writes silver_patients.
03	silver_appointments	Silver	Cleanse Appointments: date validation, join with patient, no-show flag. Writes silver_appointments.
04	silver_billing	Silver	Cleanse Billing & Claims: charge validation, claim status enrichment. Writes silver_billing.
05	silver_lab_meds	Silver	Cleanse Lab Results + Medications: abnormal flag logic, inventory calc. Writes silver_lab, silver_medications.
06	gold_kpis	Gold	Compute all 8 KPIs from Silver tables. Writes to gold_kpi_summary and individual KPI Delta tables.
07	audit_report	Ops	Query audit log table to produce pipeline run summary. Display row counts, durations, failures.

8.1 Execution Order

- Step 1 — Run Notebook 00 once (setup only)
- Step 2 — Run Notebook 01 (Bronze ingest) — can be scheduled daily
- Step 3 — Run Notebooks 02–05 (Silver) after Bronze completes — can run in parallel
- Step 4 — Run Notebook 06 (Gold KPIs) after all Silver tables are ready
- Step 5 — Run Notebook 07 (Audit Report) for monitoring

9. KPI Definitions & Business Impact

KPI	Target / Threshold	Business Impact
Patient No-Show Rate	< 10%	Reduces revenue loss from empty appointment slots
Average Length of Stay (ALOS)	Benchmark by dept.	Identifies operational inefficiencies in patient flow
HCAHPS Satisfaction Score	> 85%	Directly impacts hospital reimbursement rates
Billing Accuracy Rate	> 98%	Reduces rework, improves cash flow
Insurance Claim Approval Rate	> 95%	Reduces revenue cycle delays
Medication Inventory Turnover	4–6x per year	Prevents stockouts and medication waste
Operating Cost per Patient	Below dept. budget	Drives cost efficiency initiatives
Patient Readmission Rate	< 15%	Reduces penalties and improves patient outcomes

10. Implementation Roadmap

Phase	Step	Activity	Output
Phase 1 — Setup	1	Create ADLS containers: landing, bronze, silver, gold	Storage structure ready
	2	Upload 5 source CSV files to landing zone	Raw data available
	3	Run Notebook 00 — setup config & metadata tables	Metadata + Audit log tables created
Phase 2 — Bronze	4	Configure metadata config table for all 5 sources	Config table populated
	5	Run Notebook 01 — Bronze ingest all 5 sources	5 Bronze Delta tables + audit entries
Phase 3 — Silver	6	Run Notebooks 02–05 (can run in parallel)	5 Silver Delta tables with clean data
	7	Validate Silver tables — row counts, DQ scores	Data quality report
Phase 4 — Gold	8	Run Notebook 06 — compute all 8 KPIs	Gold KPI Delta tables ready
	9	Connect Databricks SQL / Power BI to Gold tables	Live KPI dashboard
Phase 5 — Ops	10	Schedule pipeline with Databricks Workflows (daily)	Automated daily refresh
	11	Run Notebook 07 for audit report	Audit trail dashboard

11. HIPAA Compliance & Data Governance

Healthcare data is among the most regulated in the world. This architecture is designed with HIPAA compliance in mind:

Control	Implementation	HIPAA Reference
Data Masking	PII fields (SSN, DOB, Phone) masked in Silver using SHA-256 hashing	45 CFR §164.514
Role-Based Access	Bronze restricted to data engineers; Silver/Gold has controlled access	45 CFR §164.312
Audit Log	Every data access and modification is tracked (immutable Delta table)	45 CFR §164.312(b)
ACID Transactions	Delta Lake ensures no partial writes or data corruption	45 CFR §164.306
Retention Policy	Bronze data retained for 7 years per HIPAA requirement	45 CFR §164.530(j)
No PHI in Gold	Gold tables contain only aggregated, de-identified metrics	45 CFR §164.514(b)
Column-Level Security	Unity Catalog fine-grained access control for sensitive columns	45 CFR §164.312(a)
Encryption at Rest	Azure Storage Service Encryption (AES-256) on ADLS Gen2	45 CFR §164.312(a)(2)
Encryption in Transit	TLS 1.2+ enforced for all data movement between services	45 CFR §164.312(e)
Incident Response	Pipeline failure alerts notify compliance officer within 1 hour	45 CFR §164.308(a)(6)

Note: Unity Catalog in Databricks can be used for fine-grained column-level access control for full HIPAA readiness.

12. Glossary

Term / Abbreviation	Definition
ADLS Gen2	Azure Data Lake Storage Generation 2 — scalable cloud storage on Azure
ACID	Atomicity, Consistency, Isolation, Durability — transaction guarantees in Delta Lake
Batch ID	Unique identifier assigned to each pipeline execution run
Bronze	First Medallion layer — raw, unmodified data ingestion zone
DBFS	Databricks File System — abstraction layer over cloud object storage
Delta Lake	Open-source storage format providing ACID transactions on data lakes
DQ	Data Quality — measures of accuracy, completeness, and consistency

Term / Abbreviation	Definition
EHR	Electronic Health Record — digital version of a patient’s medical history
Gold	Third Medallion layer — aggregated KPI and analytics serving layer
HIPAA	Health Insurance Portability and Accountability Act — US healthcare data regulation
HITECH	Health Information Technology for Economic and Clinical Health Act
KPI	Key Performance Indicator — measurable value demonstrating business effectiveness
Medallion	A layered data architecture pattern (Bronze → Silver → Gold)
PDD	Project Design Document — this document
PHI	Protected Health Information — individually identifiable health data regulated by HIPAA
PII	Personally Identifiable Information (e.g., SSN, DOB, address)
RBAC	Role-Based Access Control — restricting access based on user roles
SCD Type 2	Slowly Changing Dimension Type 2 — tracks historical changes with effective dates
Silver	Second Medallion layer — cleansed, enriched, and validated data
SLA	Service Level Agreement — commitment on data freshness and availability
Unity Catalog	Databricks’ governance layer for fine-grained data access control
UUID	Universally Unique Identifier — 128-bit identifier used for audit IDs